

PathScribe AI · ForMedrix

v0.9.0

PathScribe by ForMedrix AI

Admin Guide

System Administration & Configuration

For Administrators and Lead Pathologists

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Administrators access PathScribe from the Home screen via the Configuration tile, divided into four tabs: System, Staff, Protocols, and (when HL7/FHIR is active) LIS Integration.

Important: The Admin role provides configuration access only. Admins cannot see clinical case data by design. A user who needs both clinical and admin access must hold both the Pathologist and Admin roles simultaneously.

PathScribe operates in two deployment modes:

Mode	Description	Typical customer
Co-Pilot	AI assistance and synoptic reporting alongside your existing LIS. The LIS owns case creation and status.	Hospital labs running CoPath, Epic Beaker, Sunquest, etc.
Standalone LIS	PathScribe is the system of record — case management, gross/micro entry, report assembly, sign-out, and full coding.	Independent pathology groups without existing AP-LIS.

PathScribe distinguishes two categories of people — Staff (internal users who log in) and Physicians (external ordering doctors who never log in).

Category	Who they are	Log in?	Managed in
Staff	Pathologists, Residents, Administrators	Yes — login credentials	Configuration → Staff
Physicians	External ordering / referring doctors	Never — directory only	Configuration → System → Physicians

2.1 Adding a New Staff Member

- Navigate to Configuration → Staff and click Add User.
- Complete all required fields (see table below). Click Save.
- The new user can log in immediately with their email address.

Field	Required	Notes
First & Last Name	Yes	Appears on reports, audit logs, and in-app display.
Email	Yes	Used for login (if SSO not configured) and notifications.
Role(s)	Yes	One or more: Pathologist, Resident, Admin. Permissions are the union of all assigned roles.
NPI Number	No	Required for billing-related workflows.
License Number	No	State/jurisdiction medical licence.
Department	No	e.g. Surgical Pathology, Hematopathology.
Phone	No	Used for directory and urgent notification routing.
Voice Profile	Yes	EN-US or EN-GB — drives voice command recognition accuracy.
Paediatric Access	No	Enable to allow this user to see paediatric cases. Disabled by default.
Status	Yes	Active / Inactive. Inactive users cannot log in.

2.2 Roles and Permissions

Roles are not mutually exclusive. Permissions are the union of all assigned roles.

Role	Case worklist	Sign-out	Configuration	Notes
Pathologist	Full access	Primary sign-out	None	Core clinical role. Full case lifecycle.
Resident	Full access	Co-sign only	None	Sign-out gating controlled by usePermission config.
Admin	None	None	Full access	Cannot access clinical case data by design.

2.3 Deactivating and Reactivating Users

Click a user row to expand their record, then click Deactivate. The account is preserved for audit purposes but login is blocked. Click Reactivate to restore access.

2.4 Paediatric Access Control

Cases involving patients below the configured paediatric age threshold are only visible to users with Paediatric Access enabled. Users without this flag see the case row with all PHI redacted.

Navigate to Configuration → System → Physicians. Physicians appear on reports and drive result delivery routing. They do not log in and hold no permissions.

Field	Required	Notes
First & Last Name	Yes	Used on reports and result notifications.
NPI Number	Recommended	Used for billing validation and deduplication.
Fax / Email	At least one	Required for result delivery.
Preferred Contact	No	Drives automatic routing: Email, Fax, or Portal.
Client Affiliations	No	Links physician to one or more submitting facilities.
Status	Yes	Active / Inactive / Unverified. Unverified physicians do not receive notifications.

Important: Unverified physicians will not receive result notifications until an admin adds a contact method and sets the record to Active. Check the Physicians table regularly for the amber Unverified indicator.

4.1 Built-in Roles

Role	Case access	Config access	Can be deleted	Description
Pathologist	Yes	No	No	Full clinical case access and sign-out authority.
Resident	Yes	No	No	Case access with co-sign only. Sign-out gated by attending countersign.
Admin	No	Yes	No	System configuration. No clinical access.
Physician	No	No	No	External directory only. No app access.

4.2 Custom Roles

- Click Add Role. Enter a name, description, and set Case Access and Configuration Access toggles.
- Custom roles can be fully edited and deleted. Built-in roles can only have their permissions adjusted.

4.3 Permissions

Each role has a granular permissions list. Click the Permissions tab on a role record to view and edit individual action permissions (e.g. "Can finalise reports", "Can delegate cases", "Can view AI suggestions"). Permissions are enforced throughout the application.

Navigate to Configuration → System → Flags. Flags are visual markers on cases and specimens that communicate clinical context or required actions.

5.1 Case Flags vs Specimen Flags

- **Case flags** apply to an entire case (e.g. "Tumor Board — Thu 14:00", "QI Review Required", "Teaching Case").
- **Specimen flags** apply to a specific specimen within a case (e.g. "ER/PR/HER2 Pending", "BRAF Mutation", "Molecular Panel").

5.2 Creating a Flag

- Click Add Flag. Enter a name, description, colour, and severity level (1 = informational, 5 = critical).
- Set the scope: Case flag or Specimen flag.
- For specimen flags, enter the LIS Code — used by your LIS to trigger this flag automatically on result receipt.

5.3 Computational Flags

Data Source Type	Description	Required fields
LIS Endpoint	Direct REST call from PathScribe to the LIS result endpoint.	Source ID, API Endpoint URL.
AI Extraction	AI proxy parses unstructured LIS text to extract the result.	Extraction endpoint URL.
PathScribe Entry	Pathologist enters the result directly in the synoptic form.	None — no endpoint required.

5.4 Delegation Types

Navigate to Configuration → System → Delegation Types. Click + New Type, enter a Label, Description, CPT Hint, accent colour, and Active toggle. Click Create. The type immediately appears in the Delegate Modal for all active users.

Setting	Default	Description
AI Provider	Anthropic	Select Anthropic Claude or Google Gemini as the active AI provider.
Model	claude-sonnet-4-20250514	The specific model version. Claude Sonnet 4 recommended for clinical accuracy.
API Mode	Proxy	Proxy routes calls through the backend (production). Direct is development only.
Confidence Threshold	75%	Fields below this threshold show "AI: not found" instead of a suggestion. Controls which fields appear in the AI Triage modal at finalisation.
Gross-Driven AI	On	AI analyses the gross description and pre-populates synoptic fields automatically.
Microscopic-Driven AI	On	AI re-analyses after microscopic text is entered, refining suggestions and evaluating protocol assignments (see Section 9.2, Phase 3).
Auto-insert suggestions	Off	If on, AI suggestions are applied to fields without requiring pathologist confirmation.
Voice AI	Gemini active	Enable Gemini-powered voice command recognition.

Important: Changes to AI settings take effect for new sessions. Active users must log out and back in to see updated AI behaviour.

Navigate to Configuration → System. The left sidebar gives access to all system-wide settings.

7.1 LIS Integration

Setting	Default	Description
LIS Integration Enabled	Off	Master switch. Enable only when the HL7/FHIR endpoint is configured and tested.
LIS Endpoint URL	(empty)	The HL7 or FHIR endpoint for PathScribe outbound synoptic delivery.
LIS Owns Case Statuses	On	Case status is read-only in PathScribe — the LIS is authoritative. Turn off only in Standalone mode.
Allow Post-Final Actions	On	Allows addenda and amendments after sign-out. Disable if your LIS controls this workflow.

7.2 Jurisdiction

Jurisdiction	SNOMED release	ICD variant
US	SNOMED CT US Edition	ICD-10-CM
CA (Canada)	SNOMED CT CA Edition	ICD-10-CA
GB_EW (England & Wales)	SNOMED CT UK Edition	ICD-10 (WHO)
GB_SCT (Scotland)	SNOMED CT UK Edition	ICD-10 (WHO)
IE (Ireland)	SNOMED CT IE Edition	ICD-10 (WHO)

7.3 Identifier Formats

- Go to Configuration → System → Identifier Formats.
- Enter the regex pattern for your accession number format. The Example field lets you test it live.
- Enter the regex pattern for your MRN format and test similarly. Click Save.

7.4 Specimens

The specimen dictionary drives autocomplete in Case Search and case creation. Add, edit, or deactivate specimen types here.

7.5 Subspecialties

Define subspecialty pools for the delegation workflow. Cases can be routed to a subspecialty pool rather than an individual pathologist.

7.6 Terminology Services

Navigate to Configuration → System → Terminology Services. Each service shows a status badge (Connected / Degraded / Offline) and last-checked timestamp. A green badge is required before go-live.

Key architecture: A Part is an independently-versioned module (header, body section, footer). A Template is an ordered assembly of Parts. Updating a Part instantly updates every template that references it. PathScribe ships with 13 pre-built CAP-compliant Parts.

8.1 Accessing the Template Builder

Navigate to Configuration → Template Builder. The builder has two tabs: Part Library and Templates.

8.2 Creating a Part

- Click New Part in the Part Library tab.
- Enter a name, description, part type (Header / Body / Footer), specialty, and optional subspecialty.
- Set the Standard (CAP, RCPATH, or Custom). Click Create to open the Part Builder canvas.

8.3 The Canvas and Palette

Component	Description
Section	Collapsible group of related fields. Use for logical clinical groupings.
Text Field	Single-line field bound to a case data key. Renders as "Label: Value".
Data Paragraph	Multi-line text slot filled from case data. Rich text supported.
Rich Text Block	Static text identical in every report — disclaimers, institutional statements.
Dropdown	Selection field with predefined options. Single or multi-select.
Number + Unit	Numeric input with unit label. Configurable min/max and decimal places.
Date	Date field. Configurable format. Can default to today.
Computed	Calculated field derived from an expression combining other fields.
Static Label	Heading or label at any typographic level (H1–H3, body, caption).
Repeat Group	Container that iterates over a collection (e.g. specimens).
Column Layout	Arranges children in 2, 3, or 4 flowing columns.
If Block	Conditional container — children shown only when a condition is true.
Page Break	Forces a page break in the printed report.
Header	Institutional letterhead. Configure elements and scope (page 1 / all pages).
Footer	Page footer with page numbers and confidentiality notice.

8.4 Canvas Interactions

- **Add:** Drag from Palette to canvas. Drop between cards to insert at position.
- **Reorder:** Drag any card up or down within its container.
- **Resize columns:** Drag the right edge to change column span (1–12).

- **Nest:** Drag a component into a Section, Repeat Group, If Block, or Column Layout.
- **Delete:** Hover over a card and click the red trash icon.

8.5 The Inspector Panel

The Inspector shows all configurable properties for the selected component: General (Label, Width, Required, Hide if Empty), component-specific settings (binding key, options, validation), Label Formatting, and Visibility Condition.

8.6 AI Generation Configuration

Sections can be configured to allow AI-generated narrative content. Select a Section, scroll to AI Generation in the Inspector, toggle it on, enter a System instruction, and set Max tokens and Temperature (0.3 recommended for clinical accuracy).

8.7 Previewing a Part

Click Preview in the canvas toolbar to render the Part using mock clinical data. Use the page size and margin controls to check across different print formats.

8.8 Assembling a Template

In the Templates tab, click New Template. The assembly page has two panels: Left — Part Library (drag parts into the assembly area); Right — Assembly Canvas with role zones: Page 1 Header, Pages 2+ Header, Body, Page 1 Footer, Pages 2+ Footer.

Important: Minimum valid template: At least one Body Part is required to publish.

8.9 Publishing a Template

Click Publish when the assembly is complete. The template becomes immediately available for selection in the synoptic reporting workflow. In-progress cases retain the version that was active when they were started.

A synoptic report in PathScribe is a structured, field-by-field clinical summary attached to a specific specimen and template. One case can have many synoptic reports — one per specimen per protocol. This section describes the complete lifecycle, audit trail, and post-finalisation workflows.

9.1 Template-to-Specimen Assignment

- **AI-driven:** PathScribe proposes the most appropriate published template based on specimen description, clinical indication, and gross description. The pathologist confirms, modifies, or overrides.
- **Manual:** The pathologist selects a template via Add Synoptic in the right panel.

Note: Templates must be published in the Template Builder (Section 8) before they appear in the assignment list.

9.2 The Seven Phases

Phase 1 Case Creation & Specimen Registration

Case arrives from LIS (Co-Pilot) or manual entry (Standalone). Each specimen is registered with a label, type, and description.

Event	Audit Record
Case created / received from LIS	CASE_CREATED — timestamp, source, accession number
Specimen registered	SPECIMEN_REGISTERED — label, description, case ID

Phase 2 Gross Submission & AI Protocol Assignment

AI analyses the gross description and proposes a synoptic template per specimen. The pathologist accepts, modifies, or rejects.

Event	Audit Record
Gross description saved	GROSS_SAVED — content hash, word count, timestamp
AI template proposal	AI_PROTOCOL_PROPOSED — templateId, templateName, confidence
Template assigned	SYNOPTIC_ASSIGNED — instanceId, templateId, specimenId, assignedBy

Phase 3 Microscopic Entry & Protocol Re-Evaluation

When the microscopic description is saved, the AI re-evaluates whether the assigned template is still appropriate. If findings suggest a different protocol, a Protocol Change Review is raised.

Important: New in v0.9.0: The Protocol Change Review modal lists each proposed change with the current template, proposed replacement, AI reasoning, and confidence score. The pathologist selects which changes to apply and commits. Rejected changes are logged but not applied.

Event	Audit Record
Microscopic saved	MICRO_SAVED — content hash, timestamp
Protocol change proposed	PROTOCOL_CHANGE_PROPOSED — specimenId, currentTemplateId, proposedTemplateId, confidence
Protocol change accepted/rejected	PROTOCOL_CHANGE_ACCEPTED / REJECTED — instanceId, pathologistId

Phase 4 Synoptic Field Completion

AI Copilot pre-populates fields with confidence badges (green $\geq 85\%$, amber 50–84%, red $< 50\%$). Fields can be accepted with one click or overridden. Deferred synoptics (awaiting ancillary results) are excluded from the current sign-out and trigger an addendum workflow when completed.

Phase 5 Pre-Finalisation Review

When the pathologist clicks Finalise, PathScribe opens the Pre-Finalisation Review — a full-screen two-pane modal — before collecting credentials.

Important: New in v0.9.0: The Pre-Finalisation Review is mandatory for all sign-outs, replacing the previous direct-to-password flow.

- **Left pane:** All specimens and synoptic reports in transmission order. Drag to reorder. Exclude individual synoptics from this transmission.
- **Right pane:** Live Q&A; preview of the complete report output as it will be transmitted to the LIS, updating in real time.
- **Signing panel:** Biometric (if enrolled) or password. This constitutes the digital authorisation.

Phase 6 LIS Transmission (Co-Pilot Mode)

PathScribe transmits the finalised synoptic content to the LIS in the transmission order set in the Pre-Finalisation Review. Excluded synoptics remain as drafts for addendum. In Standalone mode, there is no outbound LIS transmission.

Phase 7 Post-Finalisation: Amendments & Addenda

- **Amendment:** A corrective change to a finalised report. Transmitted to the LIS as a corrected report.
- **Addendum:** An official addition to a finalised report. Used for deferred synoptics and supplementary information. PathScribe triggers the addendum workflow automatically when a deferred synoptic is completed.

Note: Neither amendments nor addenda overwrite audit records. Each creates new events with its own timestamp, author, and content hash.

9.3 Multi-Specimen Cases

Each specimen has its own independent synoptic report instance. Template assignment, AI suggestions, and field completion operate independently per specimen. The Pre-Finalisation Review shows all specimens together for ordering and exclusion before signing.

9.4 Resident Counter-Sign Workflow

When a Resident submits for sign-out, PathScribe routes it to their assigned attending for countersign rather than directly finalising. The attending countersigns (case proceeds to Pre-Finalisation Review) or returns it with a comment.

9.5 Audit Trail Summary

Every interaction generates an immutable, append-only audit record. Key properties: timestamp (UTC, millisecond precision), user ID and session ID, event type, affected entity IDs, content hashes, and AI metadata where applicable. Accessible at Home → System Audit. Export to CSV for compliance reporting.

Note: The audit log is designed to satisfy CAP, RCPATH, CLIA, and ISO 15189 audit requirements. Content hashes allow verification that stored reports have not been altered without exposing PHI.

PathScribe has 139 registered actions — every action available via mouse or keyboard is also available by voice. The Action Registry is at Configuration → Action Registry.

10.1 Architecture

Layer	Function
Voice Provider	Captures microphone input via the Web Speech API. Normalises transcripts.
Action Registry	Matches transcripts against registered voice triggers using exact match, learned triggers, then fuzzy scoring.
Event Dispatcher	Fires a PATHSCRIBE_ custom DOM event. Handlers respond per-page and per-modal.
Learning Layer	Records unrecognised phrases. When the user confirms what they meant, the mapping is stored permanently.

10.2 Coverage

Area	Actions
System, Navigation, Messages (AppShell)	44 actions
Worklist	30 actions
Synoptic Report and sub-components	60 actions
Search	3 actions
Configuration tab navigation	2 actions
Total	139 actions / 60 with keyboard shortcuts

10.3 Editing an Action

- Navigate to Configuration → Action Registry.
- Find the action by name or ID. Click to expand.
- Edit the keyboard shortcut or voice triggers (comma-separated phrases). Click Save — changes apply immediately.

Persistence: Edits persist across deployments. New actions added by developers are merged automatically. To reset to defaults, clear the `ps_action_registry` key from `localStorage`.

Navigate to Home → System Audit. The audit log records all significant system events — immutable and append-only.

- User login and logout events.
- Case creation, status changes, finalisation, and amendments.
- AI suggestion accept and reject events (confidence scores recorded).
- User account create, edit, and deactivate.
- Configuration changes (all settings, flag, role, template edits).
- Delegation and pool claim events.
- Computational test orders, cancellations, and result receipt.

Filter by date range, user, or event type. Click Export CSV for compliance reporting.

Click your initials badge → System Information to view: application version and build date, AI provider and model, API connectivity status, and browser/OS details for support diagnostics.

Click Copy to Clipboard to generate a formatted support report. The copy redacts the user's name and ID — safe to share with PathScribe support.

13. Demo Reset (Development / Staging Only)

Navigate to Configuration → Demo Reset. Visible in development and staging environments only.

- **Reset my data only** — restores your hospital's cases without affecting other testers.
- **Full reset** — restores all data for all users across all hospitals. Use sparingly.

All items below must be completed before a customer goes live. Do not grant production access until every item is verified.

#	Requirement	Owner
1	Anthropic API key stored in backend secrets manager (never in VITE_ env vars)	Engineering
2	Business Associate Agreement (BAA) signed with Anthropic before any real PHI is processed	Legal
3	Anthropic training data opt-out confirmed for API account	Engineering
4	Production backend proxy built for all AI calls — Vite dev proxy is development only	Engineering
5	AMA CPT data licence procured (approx. \$3–10k/year) before CPT codes are displayed	Legal
6	CPT backend proxy endpoint built at /api/terminology/cpt	Engineering
7	UMLS licence obtained for full SNOMED CT	Legal
8	SNOMED CT production endpoint configured via UMLS API with backend key injection	Engineering
9	Non-US ICD-10 variant proxy configured if customer uses RCPATH, RCPA, or ICCR protocols	Engineering
10	Firestore encryption at rest confirmed enabled in GCP project	Engineering
11	Flag definitions seeded for customer jurisdiction and lab type	Implementation
12	Governing body configured for customer region (CAP / RCPATH / RCPA / ICCR)	Implementation
13	AI provider set to Claude Sonnet 4 (recommended for clinical accuracy)	Implementation
14	All Terminology Services health checks passing (Configuration → System → Terminology Services)	Implementation
15	PHI de-identification implemented before AI prompt submission with real patient data	Engineering
16	Penetration test completed by external vendor before go-live	Security

Task	Navigation path
Add a staff user	Configuration → Staff → Add User
Deactivate a user	Configuration → Staff → [user] → Deactivate
Manage physician directory	Configuration → System → Physicians
Edit role permissions	Configuration → Staff → Roles → [role] → Permissions
Create a flag	Configuration → System → Flags → Add Flag
Configure computational flag	Configuration → System → Flags → [flag] → Computational
Add a delegation type	Configuration → System → Delegation Types → + New Type
Open Template Builder	Configuration → Template Builder
Create a new Part	Template Builder → Part Library → New Part
Assemble a template	Template Builder → Templates → New Template
Publish a template	Template Assembly page → Publish
Change AI model	Configuration → AI Settings → Model
Set confidence threshold	Configuration → AI Settings → Confidence Threshold
Configure identifier formats	Configuration → System → Identifier Formats
Check terminology services	Configuration → System → Terminology Services
View synoptic lifecycle & audit events	Admin Guide Section 9
View action registry	Configuration → Action Registry
View audit log	Home → System Audit
System information	Nav bar initials badge → System Information
Generate support report	System Information → Copy to Clipboard
Reset demo data	Configuration → Demo Reset (development only)